

MARKUS DE MEDEIROS

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EDUCATION

New York University, Courant Institute

PhD Computer Science, in progress.

New York, NY

September 2023 – Present

University of British Columbia

BSc Combined Honours in Computer Science and Mathematics.

Vancouver, BC

September 2018 – April 2023

- Stanley M. Grant Scholarship in Mathematics: 2022. By recommendation of the department.
- Dean's Honour List: 2018-2023.
- NSERC USRA award: 2021, 2023; SURE award: 2022.

EXPERIENCE

Neuron Relational Logic: Verified Kernels for Machine Learning

May 2025 – August 2025

Amazon Web Services

- Develop a separation logic for verifying systems-level machine learning programs.
- Organize an ongoing community-driven port of the Iris framework to the Lean 4 theorem prover.

Applications of Higher-Order Separation Logic in Probabilistic Verification

September 2023 – Present

with Joseph Tassarotti (NYU)

- Verifying security and privacy properties of distributed systems in Rocq.
- Expanding the functionality of the Clutch verification framework.
- **Published** Error Credits: Resourceful Reasoning about Error Bounds for Higher-Order Probabilistic Programs. ICFP (2023). DOI. **ICFP Distinguished Paper Award.**
- **Published** Tachis: Higher-Order Separation Logic with Credits for Expected Costs. OOPSLA (2024). DOI.

SampCert: Verified Differential Privacy

May 2024 – August 2024

Amazon Web Services

- Develop and deploy a discrete differential privacy library, verified using the Lean 4 theorem prover.
- Verify high-level privacy results, low-level optimizations, and develop a unifying privacy framework.
- **Published** SampCert: Verified Foundations for Differential Privacy. PLDI (2025) DOI. **PLDI Distinguished Artifact Award**

Coupled Borrows: Automated, Extensible Memory Safety Proofs for Prusti

May 2022 – August 2023

with Alex Summers (UBC), Aurel Bily (ETH Zurich)

- Design a robust semantics for the internal state of the Rust type checker as an external API for verification tools.
- Collaborate with Prusti and Rust developers to verify programs with realistic borrowing features.
- to balance their evolving design requirements.
- **BSc. Thesis** and presentation link.
- **Published** Place Capability Graphs: A General-Purpose Model of Rust's Ownership and Borrowing Guarantees. OOPSLA (2025).

Fractional Differential Equations

May 2021 – August 2021

with Jun-cheng Wei (UBC)

- Develop numerical simulations of nonlocal differential equations to corroborate our asymptotic theory.
- **Published** Existence and stability of symmetric and asymmetric patterns for the half-laplacian gierer-meinhardt system in one-dimensional domain. *Mathematical Models and Methods in Applied Sciences* (2022). DOI.
- **Published** Spike Solutions to the Supercritical Fractional Gierer-Meinhardt System. *JNS* (2024). DOI.

Teaching Assistantships

University of British Columbia and New York University

- UBC: CPSC 312 *Functional and Logic Programming*, CPSC 421 *Theory of Computation*, CPSC 310 *Software Engineering*, Math 184 *Differential Calculus*. NYU: CSCI-GA 2110 *Programming Languages*.

TECHNICAL SKILLS

Languages: Rust, OCaml, Haskell, Python, C/C++, Prolog, Java, Typescript

Verification Tools: Lean, Rocq/Coq, Iris, Prusti, Viper, Z3

Research Interests: Automated reasoning, separation logic, distributed and concurrent verification, functional programming, type systems, verified compilers, systems, differential privacy, security.